

14 For destructive interference at Q,

difference of the 2 paths starting from point P to point Q = $(2m + 1)\frac{\lambda}{2}$

$$(l_1 + l_3) - (l_2 + l_4) = (2m + 1)\frac{\lambda}{2}$$

Answer : D

15 With a constraint at mid point $l_1 - l_2 = n\frac{\lambda}{2}$ where n is even number